

SOEs as intermediation: Leakage effect under financial repression[☆]Zhiyong Yao^{a,*}, Dingwei Gu^b, Wen Cao^c^a School of Management, Fudan University, 670 Guoshun Road, Shanghai 200433, China^b School of Economics, Fudan University, China^c School of Management, Fudan University, China

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ABSTRACT

This article sheds new light on the puzzle why and how China's economy and private sector can grow so remarkably despite serious financial repression and credit control. We show that there exists a “leakage effect”: State-owned enterprises borrow from banks with the official interest rate, and then relend their loans to private firms with the (black) market interest rate. When doing so, all parties involved are better off, and the inefficiency of financial repression is mitigated.

1. Introduction

Financial repression is common in developing countries. It dampens the growth of the financial sector, which is critical for economic development (McKinnon 1973, 1993; Shaw, 1973; Roubini and Sala-i-Martin, 1992, 1995). Furthermore, the government often rations the available credit to the so-called priority sector. It thwarts economic growth through adverse effects on both the volume and the productivity of investments (World Bank, 1989).

Under the planned economy, Chinese government had suppressed financial markets entirely. After the start of economic reform in 1978, China has continued implementing financial repression policies despite occasional financial reforms. Until most recent years, Chinese government has owned and controlled the banking sector, depressed the interest rates for both deposits and loans, and channeled available credits to the “priority” state-owned enterprises (SOEs) through directed credit programs.

Meanwhile, the burgeoning domestic private sector faces serious credit constraints. For example, in years 1990 and 1994, non-state or private sector obtained a tiny 0.3% and 0.6% of total short-term loans from state banks, while SOEs received 99.7% and 96.5%, respectively.¹ As inflation risen over time, the real interest rate was negative for many of the years. Thus interest rates control is also an implicit subsidy to borrowers, which are primarily SOEs. Studies find that SOEs in China have much easier access to bank loans, and accordingly, they have higher leverage than non-state firms (Fan et al., 2006; Li et al., 2007).

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¹ Please refer to the China Financial Yearbooks 1991 and 1995, respectively.

Despite the serious financial repression, however, China has experienced remarkable economic growth for more than three decades. Even though the government has monopolized most of the financial resources and channeled them into SOEs, the productivity and the growth rate of private enterprises (PEs) are winning the race. In the past three decades, the private sector had grown at double-digit rate each year, from almost nothing to account for more than 60% of China's GDP. And SOEs' share had declined from approximately 80% in 1978 to less than 30%. Starting in 1998, the Chinese government has explicitly given up the policy that directs bank credits to SOEs. In recent years, the government has even encouraged banks to lend to the small- and the medium-size enterprises (SMEs), which are primarily PEs. Nevertheless, most available loanable funds still go to SOEs. For example, SOEs received 72% and 66.9% of total short-term loans in the years 1998 and 2001, while PEs obtained only 0.8% and 1.36%, respectively.²

In this article, we propose a model to shed new light on three questions or puzzles: How can China's economy grow so remarkably since 1978 despite serious financial repression? How can the private sector grow so fast despite most of the financial resources go to SOEs? And, why do banks continue offering most loans to SOEs, even after the government no longer imposes such a requirement?

We argue that despite financial repression, there exists a “leakage effect” between SOEs and PEs. When government controls the interest rate and/or the loan allocation, SOEs obtain funds from banks by the officially controlled interest rate. And then they “relend” their loans to the PEs by the “black” market interest rate. When doing so, banks, SOEs, and PEs are all better off. In other words, self-interested agents have found ways or created the black market transactions to bypass the inefficiency of financial repression. It can help explain the first two puzzles raised above.

For the third puzzle, there already exist several explanations, such as the political correctness argument, and the collateral argument. We offer a different hypothesis as follows. With the interest rate control but not the allocation control, banks may suffer losses lending to PEs directly. However, lending indirectly to PEs going through SOEs will not result in such losses. This is because by “unofficially” delegating the intermediation role to the SOEs, the bank shifts the risk burden to them. In other words, when SOEs relend their loans to the PEs, they actually take over the risk by paying back certain return to the bank. At the same time, their risk-taking behavior is compensated by the higher (black) market return.

Therefore, SOEs acting as financial intermediation is beneficial in two aspects: First, it helps reallocate scarce financial resources to the private sector³; second, it helps reduce the bank's risks.

There are many kinds of literature trying to understand China's economic growth and the private sector's development despite financial repression. Allen et al. (2005, 2008) suggest that there exist informal financing channels and governance mechanisms that support the fast growth of the private sector in China. Tsai (2002) surveys private entrepreneurs in three regions of China, and compiles information on informal credit channels to private business. In this article, we identify a new financing channel: SOEs relending their bank loans to PEs, which is critical but barely mentioned in the previous literature. And the relending activities actually combine both the formal and the informal financing.

Using Chinese provincial data, Lu and Yao (2009) find that the enhanced legal system suppresses private investment. They interpret their finding as the evidence for a leakage process that moves financial resources from the state sector to the private sector. They point out several channels of leakage, such as the trade credit, the diversion of assets and bank credits, and SOEs restructuring and privatization.⁴ Our paper further elaborates and formalizes the “leakage effect” mentioned by Lu and Yao (2009).

Our argument is also in line with the dual-track transition approach (Lau et al., 2000). But different from Lau et al. (2000), our market track for loans can be either legal or illegal, while the market track in their paper is part of the liberalization reform and thus legal. In this sense, we have broadened the dual-track approach, by including the illegal black market. Furthermore, we provide a dual-track example in the financial market in addition to those in the product and the labor markets discussed by Lau et al. (2000).⁵

As mentioned above, there are alternative explanations about why banks continue lending to SOEs rather than PEs, after the government has given up the allocation control explicitly. The political correctness argument reveals that a default by an SOE is tolerable as the SOE serves political goals, while a default by a PE will raise the concern that whether the loan officer has received a bribe from the firm (Brandt and Zhu, 2000). Another well-accepted explanation is the collateral argument. Lack of information about the riskiness of the borrowers, banks have to base their loans on the firm size and collateral. The SOEs are usually large and have enough collateral, while the PEs are usually SMEs, and thus cannot provide enough collateral (Cull and Xu, 2000).

In our model, however, we assume away the size and the collateral differences between SOEs and PEs. We exploit such a setup to examine the bank's and the SOEs' incentives when government rations credits with different instruments of financial repression. We show that even when there do not exist the size difference and the allocation control, it is still rational for the bank to choose *not* to

² Please refer to the China Financial Yearbooks 1999 and 2002, respectively.

³ Both Dollar and Wei (2007) and Liu and Siu (2007) have found a much higher return to capital in the non-state sector than in the state sector, and thus the capital reallocation from the state sector to the private sector has generated more growth in the economy.

⁴ For example, for the trade credit channel, Ge and Qiu (2005) find that the private sector is a net receiver of trade credit. Cull et al. (2006) find that SOEs with more bank credit are more likely to be suppliers of trade credit. For the bank loan diversion, Garnaut et al. (2003) find that there has been a consistent trend of rerouting of bank credits from the inland provinces, which are dominated by SOEs, to the coastal provinces, which have more PEs. For SOEs restructuring and privatization, Garnaut et al. (2003) observe that one common practice is that the old SOE spins off a new private firm, where most of the productive assets of the old SOE were transferred to the new firm.

⁵ However, there are pieces of literature with different views. For example, Li (2001) argues that financial repression has helped China maintain the macroeconomic stability needed for reform, which is perhaps more important than improving the efficiency of the financial sector. But he also recognizes that the repression has an increasing and large cost. When discussing China's growth pattern, Song et al. (2011) assume that there is no capital reallocation between the state sector and the non-state sector. It would be interesting to see whether their results are still robust or not by introducing the leakage effect into their model.

lend to PEs directly.

The rest of the paper is organized as follows. In Section 2, we introduce the institutional background in China and present some empirical evidence on relending activities. In section 3, we set up a simple model with a monopoly bank lending to different types of firms with different ownership structures. Section 4 analyzes the welfare effects under financial repression and relending activities. Section 5 concludes.

2. Institutional background and empirical evidence

2.1. Institutional background

Despite more than three decades economic reform, Chinese economy is still characterized by key instruments of financial repression until recently: regulated interest rates, state-directed credit programs, and tightly controlled capital account.⁶

In 1984 the People's Bank of China (PBC) became the official central bank after the Industrial and Commercial Bank of China (ICBC) took over its commercial banking function. The banking industry is tightly regulated by the State Council and the PBC. All four major banks in China, i.e. the ICBC, the Agricultural Bank of China, the Bank of China, and the China Construction Bank, are state-owned. State-owned banks (SOBs) have dominated lending and other financial transactions even after market-oriented reforms began to commercialize the sector. The repressive policy is mainly implemented through directed lending and interest rate control. For the purpose of this paper, we will focus on these two instruments as both of them affect investments significantly.

The main instrument for the directed-lending in China was the credit plan. In 1980s, it came in the form of the direct grant to SOEs instead of loans. The main purpose was to support the state sector which had employed most urban working forces in China. Between 1985 and 1994, several important steps were taken to further commercialize the banking system. First, the interest-bearing loans replaced the direct grants in order to harden the budget constraints of SOEs. Second, each SOB had an aggregate credit ceiling. Within those credit ceilings, the banks gained increasing autonomy to make credit decisions.

Another important step towards a more market-based allocation of credit came with the Commercial Banking Law of 1995, which attempted to transform SOBs to operate as commercial banks with accountability for profits and losses. It tried to ensure the extension of loans based on economic and not political criteria. In 1998 an indicative, non-binding target replaced the credit plan for working capital loans and fixed investment loans.

Even though banks are in general able to extend loans based on commercial considerations under the existing regulations, the government is still able to intervene to some extent in credit decision even today. For example, the Commercial Banking Law stipulates that SOBs have to provide credit for projects approved by the State Council. More subtly, the central and the local governments may have implicitly treated deposits in SOBs as inexpensive resources to supplement their fiscal needs. For example, when facing the challenge of using collective payroll taxes to pay pensions for retirees of SOEs, it is very tempting for local governments to instruct the SOEs to pay pensions on their own. In return, local governments may use their influences on local branches of SOBs to offer cheap loans to the SOEs.

Through explicit and implicit grants or credit plans, most financial resources are channeled to SOEs. Table 1 lists the breakdown of short-term loan allocation among different economy sectors during the period of 1990–2001. It shows that PEs obtained a tiny 0.3% (4.02 billion RMB) of total short-term loans, while SOEs received 99.7% (1404.23 billion RMB) in 1990. In 2001, the corresponding numbers changed to 1.36% (91.8 billion RMB) and 66.9% (4501.12 billion RMB), respectively. So before 1995, almost all of the short-term loans go to SOEs; after 1995 commercializing of the banking sector, PEs, however, barely enjoy any benefit of increased availability of financial resources to the non-state sector.

Table 2 confirms that share of total loans to the private sector is also negligibly small. Throughout the period from 1994 to 2009, the highest percentage of total loans to PEs is only 1.78%, with a minimum share of only 0.39%. The share of the non-state sector, including the TVEs (township and village enterprises), the PEs, and the Sino-foreign JVs (joint ventures), fell short of 10% for almost every year except 1999. Tables 1 and 2 demonstrate that private sector is heavily credit rationed under financial repression.

In sharp contrast to the credit rationing towards PEs are its high growth rate and increasing contribution to China's GDP. Table 3 gives the growth rate of industrial output for different economic sectors. For most of the years from 1981 to 1999, the private sector is growing at an average speed which at least triples that of the state sector.

Table 4 lists various sectors' contribution to the industrial output. After years of gradual transition, SOEs' share in China's total industrial output had declined year after year from 64.86% in 1985 to 25.52% in 1997. Meanwhile, the PEs' contribution to industrial output, had increased from 3.06% to 36.36%.⁷ The drastic change regarding the private sector's weight in Chinese economy, from non-existence before 1978 to surpassing the once dominated SOEs for the first time in the year 1996, suggests that China's repressive financial policies have not jeopardized the private sector's performance and growth. On the contrary, as suggested by Tables 3 and 4, much of the momentum in Chinese economy actually comes from the vigorous private sector. It is then puzzling to reconcile the heavy credit rationing towards the private sector and its substantial growth.

While government's control over credit allocation is relaxed over years, the government still tightly controls interest rates, both

⁶ Abiad et al.(2008) measure 91 countries' financial policies. They show that the repressive financial policy in China is very restrictive in the 80s and 90s, with a total score no higher than two and eight respectively, out of a total of 21. For years after 2000, there is still financial repression in China as the highest total score is 10.25, less than 1/2 of the maximum total score.

⁷ The non-state sector (the TVEs and the PEs) had accounted for more than 60%.

Table 1

Constitution of loans to different sectors, year 1990–2001.

	1990	1994	1998	2001
Loans(0.1 billion Yuan)	17,680.7	40,810.1	86,524.1	112,314.7
Short-term	14,082.5	27,386.3	60,613.1	67,327.3
Enterprises solely funded by the State	14,042.3	26,438.1	43,646.8	45,011.2
Industrial enterprises	5189.5	10,481.2	17,821.5	18,636.7
Commercial enterprises	5768.5	10,190.9	19,752.4	18,563.4
Construction enterprises	671.5	1121.5	1628.7	2099.6
Agriculture enterprises	2412.8	4644.5	4444.2	5711.5
Enterprises not solely funded by the State	40.2	948.2	169,663.3	22,316.1
Rural enterprises			5580	6413
Foreign-funded enterprises		792.3	2487.5	3263.5
Private enterprises	40.2	155.9	471.6	918
Others			8427.2	11,721.6
Percentage of loans to enterprises solely funded by the state	99.7	96.5	72	66.9
Percentage of short-term loans	79.6	67.1	70.1	59.9

Source: China Financial Yearbook, 1991–2002.

Table 2

The non-state sectors' share in total loans from banks, Unit: %.

	TVEs	PEs	Sino-Foreign JVs	Total
1994	5.01	0.39	1.98	7.38
1995	4.98	0.39	1.98	7.34
1996	4.61	0.46	2.20	7.27
1997	6.72	0.52	2.52	9.76
1998	6.45	0.55	2.87	9.87
1999	6.57	0.62	3.19	10.38
2000	6.10	0.66	3.07	9.83
2001	5.71	0.82	2.91	9.43
2002	5.19	0.81	2.05	8.05
2003	4.82	0.92	1.62	7.35
2004	4.53	1.17	1.23	6.93
2005	4.06	1.12	1.01	6.19
2006	2.76	1.18	0.81	4.76
2007	2.72	1.34	0.79	4.85
2008	2.46	1.39	0.75	4.60
2009	2.26	1.78	0.55	4.59

Source: CSMAR database, China Financial Yearbook, 1991–2010.

Table 3

The growth rate of gross industrial output value for state sector and non-state sector, 1980–1999 Unit:%.

Year	SOEs	TVEs	PEs	Others
1981	2.5	9.0	134.6	31.6
1982	7.1	9.5	79.0	27.7
1983	9.4	15.5	120.6	33.9
1984	8.9	34.9	97.3	56.8
1985	12.9	32.7	90.4	39.5
1986	6.2	18.0	67.6	34.2
1987	11.3	23.2	56.6	66.4
1988	12.6	28.2	47.3	61.5
1989	3.9	9.0	21.1	39.3
1990	3.0	9.0	21.1	39.3
1991	8.6	18.4	25.3	50.1
1992	12.4	33.3	47.0	64.8
1993	5.7	35.0	66.2	92.5
1994	6.5	24.9	56.3	74.3
1995	8.2	15.2	15.4	30.2
1996	5.1	20.9	20.0	23.8
1997	1.0	10.2	15.4	30.2
1998	0.1	8.1	14.7	25.3
1999	8.8	6.0	14.4	27.6

Source: China Industry Economy Statistical Yearbook, 2002.

Table 4

Contributions to gross industrial output value: state sector versus non-state sectors, 1985–1997 Unit: %.

Year	SOEs	TVEs	PEs
1985	64.86	32.08	3.06
1986	62.27	33.51	4.21
1987	59.73	34.62	5.66
1988	56.80	36.15	7.06
1989	56.06	35.69	8.25
1990	54.60	35.62	9.77
1991	52.94	35.70	11.36
1992	48.09	38.04	13.87
1993	43.13	38.36	18.51
1994	34.07	40.87	25.06
1995	33.97	36.59	29.44
1996	28.48	39.39	32.13
1997	25.52	38.11	36.36

Source: CSMAR database, China Statistic Yearbook, 1998.

the deposit and the lending rates, throughout the whole period from 1981 to 2005.⁸ Table 5 lists the administered one year deposit and lending rates, together with corresponding CPI (consumer price index) and PPI (producer price index). It is obvious that PBC controls both the deposit and the lending interest rates in such a way that has led to a significant negative return to savings and a huge implicit subsidy to corporate borrowers as inflation risen over years. For years like 1993–1995, 2004, 2010 the real borrowing rates are in fact negative. Such implicit subsidy mainly favors SOEs, which most loans are channeled to.

As to the financial markets, China's domestic stock exchanges, the Shanghai Stock Exchange and the Shenzhen Stock Exchange, were inceptioned in 1990 and grew quickly afterward. On October 23, 2009, China launched a Nasdaq-style Growth Enterprises Market (GEM), which initially listed 28 stocks, mainly from hi-tech, electronic and pharmaceutical industries. By April 2010, the number of listed firms on the GEM reached 200. Although GEM is aimed to provide financing for SMEs and Chinese regulations and laws, including Bankruptcy law and Company law, do not prohibit the listing of PEs, the accessibility of equity markets for PEs have been much lower than that for SOEs. The main listing barrier is the enforcement of the listing standards and process. Given the stock market is only a viable financing option when PEs grow to certain sizes, it raised an interesting issue of how PEs finance their “seed” capital, the most crucial capital during a firm's life cycle.

Where and how does the private sector obtain scarce financial resources to fund investment projects that make them outperform the privileged SOEs? Traditional wisdom, for example, Song et al. (2011) argue that PEs finance themselves from personal savings. Given that most of the financial resources are monopolized by the SOBs, it is implausible that the private sector obtains funds needed *only* through self-financing to sustain such a high-speed growth. We believe a big chunk of money must come from other sources, including the leakage of bank loans from SOEs to PEs.

2.2. Relending activities

In this article, the relending activity refers to SOEs borrow from banks, and then relend their loans to PEs rather than finance their own projects.⁹ By borrowing and lending at the same time to earn the interest margin, the SOEs act like financial intermediaries. There are two main forms of relending business in China: direct relending and entrusted loans (Du et al., 2016).

Direct relending is simple and straightforward: Company A borrows from the bank, and then lends the loan to company B directly, rather than financing its own project. However, direct lending between non-financial firms are forbidden by Chinese law. Due to their illegal nature, most direct relending activities go through the black or underground financial markets. Therefore, it is difficult to find direct empirical evidence of those underground financial activities.

Du et al. (2016) study direct relending using several pieces of indirect evidence. They investigate the relationship between financial assets and liabilities of companies based on the pecking order theory, and track the trace of relending cash flow in financial statements of them. They conclude that direct relending activities are prevalent across Chinese firms, especially for the SOEs.

The anecdotal evidence on the leakage of loans from SOEs to PEs is frequently mentioned by the popular press. For example, a Sina report in September 10, 2011 mentioned three big SOEs' relending behaviors.¹⁰ One is Singapore listed Yang Zijiang Ship-

⁸ There are some attempts on relaxing control over lending rate. The initial attempt to liberalize interest rates in China started in the late 1980s, when banks were allowed to adjust lending interest rates within a certain range around the administered rate. The interest ceiling on loans to SMEs was later increased by 20% in 1998 and then by 30% in 1999. From 2002, commercial banks received flexibility in setting interest rates on all loans up to 1.3 times the benchmark lending rate, which was further increased to 1.7 times in the year 2004. In October 2004, PBC changed its control over interest rate by setting an interest ceiling for deposit rate, and an interest floor for the lending rate, which is 90% of the administered rates. In 2012, the interest rate ceiling for deposit rate was increased to 1.1 times and interest rate floor for lending rate was lowered to 70% of the administered rate, respectively. PBC announced that China fully deregulates the lending interest rate control since July 20th, 2013.

⁹ SOEs can also borrow from other channels such as issuing bonds and then relend.

¹⁰ <http://finance.sina.com.cn/roll/20110910/012110463411.shtml>

Table 5
Official one year deposit and lending rate, 1991–2011.

	One Year Deposit Rate			One Year Lending Rate		
	Benchmark Deposit Rate	CPI	Real Deposit Rate	Benchmark Lending Rate	PPI	Real Lending Rate
1991	7.89	103.4	4.49	8.64	106.2	2.44
1992	7.56	106.4	1.16	8.64	106.8	1.84
1993	9.43	114.7	−5.27	9.86	124	−14.14
1994	10.98	124.1	−13.12	10.98	119.5	−8.52
1995	10.98	117.1	−6.12	11.52	114.9	−3.38
1996	9.17	108.3	0.87	11.02	102.9	8.12
1997	7.13	102.8	4.33	9.81	99.7	10.11
1998	5.03	99.2	5.83	7.55	95.9	11.65
1999	2.92	98.6	4.32	6.09	97.6	8.49
2000	2.25	100.4	1.85	5.85	102.8	3.05
2001	2.25	100.7	1.55	5.85	98.7	7.15
2002	2.02	99.2	2.82	5.39	97.8	7.59
2003	1.98	101.2	0.78	5.31	102.3	3.01
2004	2.03	103.9	−1.87	5.36	106.1	−0.74
2005	2.25	101.8	0.45	5.58	104.9	0.68
2006	2.35	101.5	0.85	5.78	103	2.78
2007	3.20	104.8	−1.60	6.71	103.1	3.61
2008	3.93	105.9	−1.97	7.18	106.9	0.28
2009	2.25	99.3	2.95	5.31	94.6	10.71
2010	2.30	103.3	−1.00	5.36	105.5	−0.14
2011	3.28	105.4	−2.12	6.34	106	0.34

Source: CSMAR database.

building Holdings, more than 25% of whose before tax profit in second quarter of 2011 came from relending to other business, instead of its core business of ship building. The report also mentioned that China Mobile had a financial subsidiary focusing on the relending; while Petro China had already set up a series of financial platforms. Also, as Reuters (Beijing, June 25th, 2013) reported, a deputy general manager from a large SOE in the steel industry revealed that the corporation did *not* use the bank loans to its own production since the average loss was about 150 RMB per ton of steel sold. Instead, his company borrowed from banks at the benchmark lending rate, and then issued loans to others at twice of the benchmark rate.

Although Chinese regulations do not allow direct lending between non-financial firms, they do allow for a financing vehicle known as entrusted loans. Entrusted loans are loans between non-financial firms using banks and other financial institutions as service agents. To make an entrusted loan, Company A (the lender) can ask a bank to “lend” A’s funds to company B (the borrower). The bank only acts as an entrust agent for a fee, and does not assume any credit risks. The interest rates of entrusted loans are negotiable using the arm’s length principle.

There are detailed regulations on entrusted loans in China. For example, as early as year 1992, PBC clarified several issues on entrusted loans, such as the definition of entrusted loans, in “A reply to the ICBC”. In 1996, PBC specified detailed rules on entrusted loans. In 2000, PBC switched the approval system to the registration system for entrusted loans.

Entrusted loans market has grown very rapidly and accumulated a huge stock.¹¹ For example, as Table 6 shows, in 2015, the growth rates of entrusted loans from the first quarter to the fourth quarter are 23.6%, 18.4%, 18.8%, and 17.2%, respectively. The growth rate is similar or even higher in the previous years. The amount of entrusted loans (stock) is 10.93 trillion RMB (about 1.7 trillion US dollars) at the fourth quarter of the year 2015, accounting for 8% of aggregate financing to the real economy (AFRE).¹²

There are two kinds of entrusted loans: the intra-corporate and the inter-corporate loans. When both the lender and the borrower are from the same business group, it is an intra-corporate loan. When the firm grants a loan to another firm which does not belong to the same business group, it is an inter-corporate loan. For the purpose of the paper, we focus on the inter-corporate entrusted loans.¹³

Table 7 gives descriptive statistics of the inter-corporate entrusted loans by SOEs that listed in Chinese Stock Exchanges for a period from 2005 to 2014. The inter-corporate loan information for listed firms are extracted from firm announcements during 2005–2012 from Resset.¹⁴ Information for the years 2013 and 2014 are downloaded from WIND. Two data sets are then merged

¹¹ The central bank collects data from banks that intermediate these loans. However, underreporting is common. Hence the numbers of the PBC statistics tend to underestimate the real size of the entrusted loans.

¹² AFRE(Stock) refers to the outstanding of financing provided by the financial system to the real economy at the end of a period, where real economy means non-financial enterprises and households.

The trust loans is another possible way which channels funds from SOEs to PEs, and the trust loans account for more than 3% of AFRE.

¹³ We fully acknowledge the fact that a lot of entrusted loans were extended to the SOEs’ subsidiaries or affiliated enterprises (HKMA, 2014). That is why we focus on the inter-corporate entrusted loans. We also acknowledge that a substantial amount of entrusted loans have flowed into sectors of overcapacity during recent years (Lu et al. 2015; Elliott et al. 2015).

¹⁴ Resset (www.resset.cn) is a widely used database for the Chinese stock market (Calomiris et al., 2010), and all listed firms are required to announce major events which may influence their stock prices, including the entrusted loans.

Table 6

Aggregate financing to the real economy (Stock). Unit: Trillion Yuan.

Items	2015.Q1		2015.Q2		2015.Q3		2015.Q4	
	Stock	Growth Rate(%)	Stock	Growth Rate(%)	Stock	Growth Rate(%)	Stock	Growth Rate(%)
AFRE(stock)	127.68	13.1	131.70	12.0	134.70	12.5	138.14	12.4
RMB loans	85.09	14.0	88.07	13.8	90.48	14.0	92.75	13.9
Foreign loans	3.48	−7.2	3.50	−6.2	3.33	−7.5	3.02	−13.0
Entrusted loans	9.67	23.6	9.87	18.4	10.35	18.8	10.93	17.2
Trust loans	5.35	4.6	5.38	1.7	5.41	4.2	5.39	0.8
Undiscounted bankers' acceptances	6.96	−7.6	6.94	−10.4	6.32	−10.8	5.85	−14.8
Net financing of corporate bonds	12.17	26.0	12.72	20.1	13.47	20.4	14.63	25.1
Non-financial enterprises equity financing	3.94	13.7	4.16	17.3	4.30	17.4	4.53	20.2

Source: Statistics and Analysis Department, PBC.

Foreign currency-denominated loans (RMB equivalent).

Equity financing on the domestic stock market by non-financial enterprises.

Table 7

Descriptive statistics of entrusted loans of SOEs listed in stock exchanges.

		2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	Total
Number of Loans		2	5	3	11	13	15	17	14	27	22	129
Loan	Mean	10,150	13,800	19,166	5796	8711	7140	7891	22,893	15,157	9610	11,676
Amount	SD	13,930	20,315	11,273	3110	7062	7753	7122	52,230	30,074	9171	23,153
(10,000 RMB)	Min	300	2000	7500	1800	2500	2000	400	700	480	500	300
	Max	20,000	50,000	30,000	10,000	24,000	30,000	30,000	200,000	150,000	40,000	200,000
Interest	Mean	10.01	10.53	6.41	13.61	9.14	15.07	14.76	11.28	9.18	9.28	11.22
Rate	SD	7.06	4.43	0.77	5.70	4.20	4.43	5.46	5.73	4.50	3.39	5.18
(%)	Min	5.02	5.81	5.83	6.11	5.31	5.40	4.82	3.00	4.00	5.60	3.00
	Max	15.00	15.00	7.29	20.00	18.00	20.00	24.50	19.20	24.00	18.00	24.50
Loan	Mean	10.5	14.4	12.0	11.9	15.2	13.0	11.6	15.4	16.1	17.1	14.6
Length	SD	2.1	5.4	0.0	6.4	8.3	6.6	2.7	6.9	14.4	16.6	10.6
(Months)	Min	9.0	12.0	12.0	6.0	3.0	6.0	5.0	12.0	6.0	1.0	1.0
	Max	12.0	24.0	12.0	24.0	36.0	36.0	18.0	36.0	84.0	84.0	84.0

together by listed company's unique stock code. Then a total of 129 entrusted loans by SOEs to unrelated cooperations (PEs) are filtered with corresponding loan details.

The descriptive statistics in Table 7 suggest that entrusted loans by SOEs are becoming increasingly common over the years. During 2005–2007 only single digit of such loans were undertaken each year; while the number of loans per year increased to double digits since 2008 and peaked in 2013 for 27 cases. The loan sizes vary a lot from 3 million to 2 billion RMB, with an average loan size of 0.117 billion RMB. The interest rates of these entrusted loans are typically much higher than the same period official bank lending rates as in Table 5, with the mean annual interest rate roughly around 11%. The length of entrusted loans also varies a lot, from a 1-month short-term loan up to a long-term loan of 7 years, with a mean length of about 15 months.

In sum, over the years there are more and more entrusted loans from SOEs to PEs, while the specific loan terms, i.e. length and interest rate, vary to support PEs' both short-run and long-term projects.

Table 7 covers only a very small sample of entrusted loans between SOEs and PEs. There are other studies on China's entrusted loans. For example, Allen et al. (2015) find that entrusted loans increase when official credits are tight and, therefore, is a market solution to credit shortage. He et al. (2015) also regard the entrusted loans as one type of informal financing and may support the high growth rate of China.

3. Model

Suppose that there exists an economy with a monopoly SOB and two groups of firms in terms of their ownership structures: Group *A* firms are SOEs and group *B* are PEs. Each firm has a project that needs to be financed. The project is either high-type (good) or low-type (bad), which is the firm's private information. Let $i = A, B$ denote the firm group, which is observable; and $j = H, L$ indicate the project type, which is unobservable. We thus have $ij = AH, AL, BH, BL$, a total of four segments. Assume that the proportion of high-type projects in each group is $\alpha_i \in [0, 1]$, $i = A, B$, which is common knowledge.

The rate of return for each project is governed by R , $R \in [-1, \bar{R}]$ with mean $\tilde{R}_{ij}$ and c.d.f. $F_{ij}(R)$.¹⁵ The projects in each group satisfy

¹⁵ The lower bound of the rate of return is -1 , to reflect the limited liability nature, and $\bar{R} > 0$ is the maximum rate of return attainable.

the following stochastic dominance conditions: $\tilde{R}_{iH} \geq \tilde{R}_{iL}$, and $\int_{-1}^x F_{iH}(R) dR \leq \int_{-1}^x F_{iL}(R) dR$ for any $x \in [-1, \bar{R}]$.¹⁶ In other words, the high-type project in each group is at least as good as the low-type project in terms of both the expected return and the risk.

Define $\tilde{R}_i \equiv \alpha_i \tilde{R}_{iH} + (1 - \alpha_i) \tilde{R}_{iL}$ to be the average expected return of all projects in each group; let $\phi_i \equiv \tilde{R}_{iH} - \tilde{R}_{iL} \geq 0$ and $\varphi_i(x) \equiv \int_{-1}^x F_{iL}(R) dR - \int_{-1}^x F_{iH}(R) dR \geq 0$ indicate the differences of the return and the risk between the high-type and the low-type projects in each group, respectively, where $i = A, B$. For each firm, one unit of fund is required to undertake the project. Assume that firms have no extra funds for their projects and have to borrow from the bank, and the bank has only one unit of loan to be allocated. Therefore, there is a shortage of fund supply, and only one firm would obtain the fund. Also, assume that the bank charges the same collateral for all firms, which is normalized to be zero.

As the bank is a monopoly and the ownership of a firm is observable, the bank may charge different interest rates based on the ownership of the firm, which is the third-degree price discrimination. The bank's required return for the firms in group i is $r_i = 1 + \kappa_i$, where κ_i is the interest rate, $i = A, B$. Firm ij is said to be bankrupt if $R_{ij} < r_i$. In the event of bankruptcy, the bank becomes the owner of the investment project and its returns. Thus, the expected rate of return to firm ij by undertaking the project is:

$$\begin{aligned} \pi_{ij}(r_i) &= E\{\max[(R_{ij} - r_i); 0]\} \\ &= (\tilde{R}_{ij} - r_i) + \int_{-1}^{r_i} F_{ij}(R) dR. \end{aligned} \quad (1)$$

Correspondingly, the bank receives the expected return:

$$\begin{aligned} v(r_i) &= E\{\min[r_i; R_{ij}]\} \\ &= r_i - \int_{-1}^{r_i} F_{ij}(R) dR. \end{aligned} \quad (2)$$

Define $\bar{r}_{ij}$ as the solution to $\pi_{ij}(\bar{r}_{ij}) = 0$. $\bar{r}_{ij}$ is the maximum interest rate that each type of firms can afford to pay ex ante. To facilitate the analysis, we first derive the following Lemma:

Lemma 1. Asymmetric information about the project's type leads to adverse selection if and only if $\bar{r}_{iH} < \bar{r}_{iL}$, or equivalently, $\phi_i < \varphi_i(\bar{r}_{iH})$.

Proof. Please see the appendix

If the bank charges interest rate r_i such that $\bar{r}_{iH} < r_i < \bar{r}_{iL}$, the high-type firm will drop out, but the low-type will obtain the loan. Hence, adverse selection arises.

From now on, we assume that $\bar{r}_{iH} < \bar{r}_{iL}$ (equivalently, $\phi_i < \varphi_i(\bar{r}_{iH})$), and all of our analyses below are based on this assumption. That is, there exists an adverse selection due to the severe asymmetric information problem.

4. Analysis

In this section, we analyze and compare three scenarios: (1) no government control; (2) government controlling both the interest rate and the loan allocation; (3) government controlling the interest rate only.

4.1. No government control

Given $\bar{r}_{iH} < \bar{r}_{iL}$, the monopolistic bank faces a trade-off: It can charge either $\bar{r}_{iH}$ so that both the high- and the low-type firms are able to apply for the loan, or $\bar{r}_{iL}$, and only the low-type firms can afford to borrow. By comparing the bank's expected returns under these two situations, we have the following Lemma:

Lemma 2. The bank will charge $\bar{r}_{iH}$ and make both type firms willing to apply the loan if $\phi_i > (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$; the bank will charge $\bar{r}_{iL}$ and only serve low-type firms if $\phi_i \leq (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$.

Proof. Please see the appendix.

When $\phi_i \leq (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$, or the return difference between the high-type and the low-type is small enough, the bank will only serve low-type firms by charging $\bar{r}_{iL}$. It will serve the low-type PE if $\tilde{R}_{AL} < \tilde{R}_{BL}$; otherwise, it will serve the low-type SOE. $\tilde{R}_{AL}$ and $\tilde{R}_{BL}$ are also the social welfare of serving low-type SOEs and PEs, respectively. Obviously, the bank's decision to lend to the low-type firms is social inefficient.

When $\phi_i > (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$, or the return difference between the high-type and the low-type is large enough, the bank is willing to charge lower interest rate $\bar{r}_{iH}$ to serve both types. The social welfare is either $\tilde{R}_A$ or $\tilde{R}_B$, depending on which group the bank finances, and the bank obtains return of $\tilde{R}_{iH} - (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$. The bank is willing to serve SOEs if

¹⁶ Our assumption contains both the first-order and the second-order stochastic dominances. If $\tilde{R}_{iH} > \tilde{R}_{iL}$, it is the first-order stochastic dominance. If $\tilde{R}_{iH} = \tilde{R}_{iL}$, it is the second-order stochastic dominance.

$\tilde{R}_{AH} - (1 - \alpha_A)\varphi_A(\bar{F}_{AH}) > \tilde{R}_{BH} - (1 - \alpha_B)\varphi_B(\bar{F}_{BH})$, which is equivalent to

$$\underbrace{\hat{R}_B - \hat{R}_A}_{\text{Welfare difference}} < \underbrace{(1 - \alpha_B)[\varphi_B(\bar{F}_{BH}) - \phi_B]}_{\text{Information rent to flow-type PEs (+)}} - \underbrace{(1 - \alpha_A)[\varphi_A(\bar{F}_{AH}) - \phi_A]}_{\text{Information rent to flow-type SOEs (+)}} \equiv X. \quad (3)$$

Notice that $\hat{R}_B - \hat{R}_A$ characterizes the *welfare difference* between PEs and SOEs, and $[\varphi_i(\bar{r}_{iH}) - \phi_i]$ represents the low-type firms' profit. Given that the bank charges low interest rate $r_l = \bar{r}_{iH}$, the high-type firms are break even, but the low-type firms are able to obtain positive profit, which can be seen as *information rent* due to the asymmetric information. Since the proportion of low-type firms is $(1 - \alpha_i)$, the information rent for low-type firms in each group is $(1 - \alpha_i)[\varphi_i(\bar{r}_{iH}) - \phi_i]$. Then X represents the *information rent difference* between the low-type PEs and SOEs.

When $\hat{R}_B - \hat{R}_A < X$, or the welfare difference is smaller than the information rent difference (or equivalently, the SOEs bring the bank higher expected return than the PEs), the bank will offer the loan to the SOEs, and the social welfare will be $\hat{R}_A$. Otherwise, when $\hat{R}_B - \hat{R}_A \geq X$, the bank will lend to the PEs, and the social welfare will be $\hat{R}_B$.¹⁷ Whether these allocations are efficient or not depends on the comparison between $\hat{R}_B$ and $\hat{R}_A$.

In summary, we have the following Proposition:

Proposition 1. Suppose that there is no government control.

1. Given $\phi_i \leq (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$, the bank will lend to the low-type firms, the social welfare is $\max\{\hat{R}_{AL}, \hat{R}_{BL}\}$, and the allocation is not efficient.
2. Given $\phi_i > (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$, there are four cases: (a) When $0 < \hat{R}_B - \hat{R}_A < X$, the loan will be offered to the SOE; (b) when $X \leq \hat{R}_B - \hat{R}_A < 0$, the bank will lend to the PE. In both (a) and (b), the allocations are inefficient. (c) When $\hat{R}_B - \hat{R}_A \geq \max\{0, X\}$, the bank will lend to the PE; (d) when $\hat{R}_B - \hat{R}_A \leq \min\{0, X\}$, the bank will lend to the SOE. In both (c) and (d), the allocations are efficient.

The intuition is straightforward. When $\phi_i \leq (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$, the bank only serve low-type firms, which is standard adverse selection problem and no doubt is inefficient. When $\phi_i > (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$, the bank can serve both types of firms in either SOEs or PEs groups. But which group will be preferred depends on the comparison of the expected return to the bank across groups. The bank's private interest may or may not align with the social interest. In the above Proposition, we characterize four cases. In (c) and (d), the profit maximization of the bank coincides with the maximization of the social welfare and leads to the efficient allocations; while in (a) and (b), they do not.¹⁸

4.2. Both interest rate and allocation controls

Consider the case that the government dictates the bank to finance SOEs, and meanwhile controls the interest rate $r_A \leq \bar{F}_{AH}$ so that the SOEs can afford to borrow. In the following analysis, we will discuss two episodes: (1) without relending; (2) with relending.¹⁹

4.2.1. Without relending

Since both interest rate and allocation are under control, the bank has to finance SOEs with interest rate $r_A \leq \bar{F}_{AH}$. The bank will charge $r_A = \bar{F}_{AH}$ to make both type SOEs able to apply the fund, and achieve return of $\tilde{R}_{AH} - (1 - \alpha_A)\varphi_A(\bar{F}_{AH})$. The social welfare is $\hat{R}_A$. We compared it with the above no-control scenario to see whether control improves social welfare or not.

In the no-control scenario discussed in section 4.1, when $\phi_i \leq (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$, the bank will serve low-type SOEs if $\tilde{R}_{BL} < \tilde{R}_{AL}$, which is equivalent to $\hat{R}_B - \hat{R}_A \leq \alpha_B\phi_B - \alpha_A\phi_A$. The interest-rate control makes the bank can also serve high-type SOEs. Thus it will no doubt improve the social welfare. If $\hat{R}_B - \hat{R}_A > \alpha_B\phi_B - \alpha_A\phi_A$, the bank will only serve low-type PEs in the no-control case. The government control can improve social welfare if $\hat{R}_A > \tilde{R}_{BL}$, which is equivalent to $\hat{R}_B - \hat{R}_A < \alpha_B\phi_B$. However, when $\hat{R}_B - \hat{R}_A \geq \alpha_B\phi_B$, government control can only lead to welfare loss.

When $\phi_i > (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$, the bank will serve both types of SOEs anyway if $\hat{R}_B - \hat{R}_A < X$, as cases (a) and (d) in Proposition 1 indicate. So the government control has no effect on the allocation as well as the social welfare. But when $\hat{R}_B - \hat{R}_A > X$, in the no-control case the bank will serve both types of PEs. The government control will improve social welfare when $X \leq \hat{R}_B - \hat{R}_A < 0$ (case (b) in Proposition 1). However, when $\hat{R}_B - \hat{R}_A \geq \max\{0, X\}$ (case (c) in Proposition 1), government control results in welfare loss.

In summary, we have the following Proposition:

Proposition 2. Suppose that there are both interest rate and allocation controls.

1. Given $\phi_i \leq (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$, allocation and interest rate controls improve social welfare when $\hat{R}_B - \hat{R}_A < \alpha_B\phi_B$, but reduce it when $\hat{R}_B - \hat{R}_A \geq \alpha_B\phi_B$.
2. Given $\phi_i > (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$, allocation and interest rate controls have no effect on social welfare when $\hat{R}_B - \hat{R}_A < X$, and reduce it when $\hat{R}_B - \hat{R}_A \geq \max\{0, X\}$. However, the controls improve social welfare when $X \leq \hat{R}_B - \hat{R}_A < 0$.

When $\phi_i \leq (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$, the no-control allocation is always social inefficient as it only serves low-type firms, as discussed in section 4.1. Only when SOEs are efficient enough ($\hat{R}_B - \hat{R}_A < \alpha_B\phi_B$), the control is Pareto improvement. Otherwise, the control leads

¹⁷ When $\hat{R}_B - \hat{R}_A = X$, the bank is indifferent between serving PEs and SOEs. To simplify the matter, we make this assumption.

¹⁸ There also exist two asymmetric cases: $\phi_A > (1 - \alpha_A)\varphi_A(\bar{F}_{AH})$ but $\phi_B < (1 - \alpha_B)\varphi_B(\bar{F}_{BH})$, and $\phi_A < (1 - \alpha_A)\varphi_A(\bar{F}_{AH})$ but $\phi_B > (1 - \alpha_B)\varphi_B(\bar{F}_{BH})$. We do not analyze them because the two cases involve too many trivial discussions without adding any new insights, which will distract our attention.

¹⁹ Relending here can be seen as resale.

to welfare loss.

When $\phi_i > (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$, as we discussed in Proposition 1, there are four cases in total. In two of them, the loan is allocated to the SOEs even under the no-control scenario, thus the controls will not change anything. In the other two cases, (b) and (c), the fund is offered to the PEs. When $\hat{R}_B - \hat{R}_A \geq \max\{0, X\}$, it is social optimal to finance PEs, thus control will no doubt reduces social welfare. When $X \leq \hat{R}_B - \hat{R}_A < 0$, the bank will lend the fund to the PEs, which is inefficient. Thus, the controls improve social welfare by directing the fund to the SOEs.

4.2.2. With relending

Given that interest rate $r_A = \bar{r}_{AH}$, the high-type SOEs achieve zero return while the low-type SOEs obtain positive return if they financing their own projects. Suppose it is the high-type SOE that obtains the loan and then relends to PEs. When $\phi_B \leq (1 - \alpha_B)\varphi_B(\bar{r}_{BH})$, the SOE (just like the bank in the no-control scenario in section 4.1) only relends to low-type PEs and achieves revenue $\tilde{R}_{BL}$. The relending will occur if:

$$\tilde{R}_{BL} - \bar{r}_{AH} = \underbrace{\tilde{R}_{BL} - \tilde{R}_{AH}}_{\text{Relending efficiency}} - \underbrace{\int_{-1}^{\bar{r}_{AH}} F_{AH}(R)dR}_{\text{Risk suffered by bank (+)}} > 0.$$

When $\phi_B > (1 - \alpha_B)\varphi_B(\bar{r}_{BH})$, the SOE charges low interest rate $\bar{r}_{BH}$ and achieves revenue $\tilde{R}_{BH} - (1 - \alpha_B)\varphi_B(\bar{r}_{BH})$. The relending will occur if:

$$\begin{aligned} & \tilde{R}_{BH} - (1 - \alpha_B)\varphi_B(\bar{r}_{BH}) - \bar{r}_{AH} \\ = & \underbrace{\hat{R}_B - \tilde{R}_{AH}}_{\text{Relending efficiency}} - \underbrace{(1 - \alpha_B)(\varphi_B(\bar{r}_{BH}) - \phi_B)}_{\text{Information rent of low-type PEs (+)}} - \underbrace{\int_{-1}^{\bar{r}_{AH}} F_{AH}(R)dR}_{\text{Risk suffered by bank (+)}} > 0. \end{aligned}$$

The above two formulas indicate the conditions of high-type SOE's willingness to relend. When the SOE only deals with the low-type PEs, its relending takes over the bank's risk ($\int_{-1}^{\bar{r}_{AH}} F_{AH}(R)dR$) by paying back certain return $\bar{r}_{AH}$. When the SOE relends to both the high- and the low-type PEs, its relending not only takes over bank's risk, but also creates information rent for the low-type PEs.

If it is the low-type SOE that obtains the loan and then relends to PEs, we can repeat the above analysis.²⁰ Therefore, we have the following Proposition:

Proposition 3. Suppose that there is both interest rate and allocation control, and SOEs can relend. Relending can always improve social welfare. The three parties involved, the bank, the SOE, and the PE, are all better off.

Relending, as long as it happens, is beneficial to all the parties involved. The relending raises the bank's return because the SOE, as an intermediary, takes over the risk by paying back certain return. Either the high-type SOE gets positive return rather than break even if engaging in relending, or the low-type SOE obtains higher return if it implements the relending. Meanwhile, either the high-type PE gets its project financed if it obtains the relending fund, or the relending may benefit the low-type PE as there is an additional information rent for it, besides financing its project. Therefore, all the three parties involved, the bank, the SOE, and the PE, are better off. Relending is Pareto improvement and can always improve social welfare.

4.3. Interest rate control only

A commonly used instrument of financial repression is the interest rate control. As mentioned in section 2, Chinese government gave up the allocation control explicitly in 1998 but still firmly controls the interest rate until recently. In this part, we examine how the interest rate control affects the fund allocation and the efficiency. Since the government still wants to support the SOEs, it is reasonable to assume that the lending rate ceiling is fixed at $\bar{r} = \bar{r}_{AH}$. One can easily infer that $\bar{r}_{BH} \geq \bar{r}_{AH}$. Otherwise, interest rate control is not necessary.

Given that there is only the interest rate control, it is up to the bank to decide the fund allocation. In the following analysis, we also discuss and compare the two situations: (1) without relending; (2) with relending.

4.3.1. Without relending

When there is no relending, by financing SOEs, the bank achieves return of $\bar{r}_{AH} - \left[\alpha_A \int_{-1}^{\bar{r}_{AH}} F_{AH}(R)dR + (1 - \alpha_A) \int_{-1}^{\bar{r}_{AH}} F_{AL}(R)dR \right]$. By lending to PEs, the return of bank can be shown as $\bar{r}_{AH} - \left[\alpha_B \int_{-1}^{\bar{r}_{AH}} F_{BH}(R)dR + (1 - \alpha_B) \int_{-1}^{\bar{r}_{AH}} F_{BL}(R)dR \right]$. Therefore, with interest rate control, the bank will serve SOEs if:

²⁰ Please see the appendix for the low-type SOE case under title “Case 1”.

$$\begin{aligned}\widehat{R}_B - \widehat{R}_A &< (1 - \alpha_B)[\varphi_B(\bar{r}_{BH}) - \phi_B] - (1 - \alpha_A)[\varphi_A(\bar{r}_{AH}) - \phi_A] + Y \\ &= X + Y,\end{aligned}\quad (4)$$

where $Y = (\bar{r}_{BH} - \bar{r}_{AH}) - \left[\alpha_B \int_{\bar{r}_{AH}}^{\bar{r}_{BH}} F_{BH}(R) dR + (1 - \alpha_B) \int_{\bar{r}_{AH}}^{\bar{r}_{BH}} F_{BL}(R) dR \right] > 0$. Compared to (3), Y indicates the PEs' additional revenue because they may obtain the loan at a lower interest rate due to interest rate control.

When (4) holds, the bank will lend to SOEs and the social welfare is $\widehat{R}_A$. Otherwise, the bank will lend to PEs and the social welfare is $\widehat{R}_B$. By comparing (4) to (3), we can conclude that the additional term makes lending PEs less attractive, compared to the government no-control scenario.

We thus have the following Proposition:

Proposition 4. Suppose that there is interest rate control and there is no relending. Compared to the no government-control scenario,

1. Interest rate control reduces bank's incentive to serve PEs, and enhances bank's incentive to serve SOEs.
2. When $(\widehat{R}_B - \widehat{R}_A) \in (X, X + Y)$, the loan is offered to SOEs rather than PEs. Interest rate control improves social welfare when $(\widehat{R}_B - \widehat{R}_A) \in (X, \min\{0, X + Y\})$, but reduces it when $(\widehat{R}_B - \widehat{R}_A) \in (\max\{0, X\}, X + Y)$. Within other parameter ranges, the interest rate control does not change the allocation and the welfare.

Proof. Please see the appendix.

The intuition is quite simple. Compared to the no control scenario, when there is interest rate control, the bank's return from financing SOEs remains the same, but its return of serving PEs decreases due to lower interest rate $\bar{r}_{AH} \leq \bar{r}_{BH}$. Therefore, the bank's incentive to serve PEs decreases. Consequently, it has a stronger incentive to serve SOEs.

When $\widehat{R}_B - \widehat{R}_A \leq X$, the loan will be offered to the SOEs; when $\widehat{R}_B - \widehat{R}_A \geq X + Y$, the bank will lend to the PEs. In both cases, the allocations are the same as the government no-control scenario. Therefore, interest rate control has no effect on the allocation and the social welfare. However, when $(\widehat{R}_B - \widehat{R}_A) \in (X, X + Y)$, the loan is offered to PEs if there is no control, but will be offered to SOEs if there is interest rate control. Under this situation, interest rate control improves social welfare when $(\widehat{R}_B - \widehat{R}_A) \in (X, \min\{0, X + Y\})$ as $\widehat{R}_B < \widehat{R}_A$, or it is social optimal to allocate the loan to SOE. It will reduce the social welfare when $(\widehat{R}_B - \widehat{R}_A) \in (\max\{0, X\}, X + Y)$ as $\widehat{R}_B > \widehat{R}_A$, since it is social optimal to allocate the loan to PEs.

4.3.2. With relending

Interest rate control reduces bank's incentive to serve PEs, and enhances bank's incentive to serve SOEs, as Proposition 4 indicates. Therefore, we focus on SOEs' incentive of relending. Notice that when $(\widehat{R}_B - \widehat{R}_A) \in (0, X + Y)$, the bank choosing to serve SOEs is social inefficient. Similar to the previous relending case, we analyze the relending incentive of high-type SOE.²¹

When $\phi_B \leq (1 - \alpha_B)\varphi_B(\bar{r}_{BH})$, the relending return can be shown as $(\widetilde{R}_{BL} - \widetilde{R}_{AH}) - \int_{-1}^{\bar{r}_{AH}} F_{AH}(R) dR$. The SOE is willing to relend if and only if $\widetilde{R}_{BL} - \widetilde{R}_{AH} > \int_{-1}^{\bar{r}_{AH}} F_{AH}(R) dR$.

When $\phi_B > (1 - \alpha_B)\varphi_B(\bar{r}_{BH})$, the relending return is $(\widehat{R}_B - \widetilde{R}_{AH}) - (1 - \alpha_B)(\varphi_B(\bar{r}_{BH}) - \phi_B) - \int_{-1}^{\bar{r}_{AH}} F_{AH}(R) dR$. Therefore, the SOE is willing to relend if and only if $\widehat{R}_B - \widetilde{R}_{AH} > (1 - \alpha_B)(\varphi_B(\bar{r}_{BH}) - \phi_B) + \int_{-1}^{\bar{r}_{AH}} F_{AH}(R) dR$.

Knowing that SOE is willing to relend, by financing SOEs, the bank's expected return becomes $\bar{r}_{AH}$. By lending to PEs, the expected return of bank can be shown as $\bar{r}_{AH} - \left[\alpha_B \int_{-1}^{\bar{r}_{AH}} F_{BH}(R) dR + (1 - \alpha_B) \int_{-1}^{\bar{r}_{AH}} F_{BL}(R) dR \right] < \bar{r}_{AH}$. Therefore, given the bank anticipating SOEs will relend, the bank has no incentive to serve PEs. Therefore, we have the following Proposition:

Proposition 5. Suppose that there is interest rate control, and SOEs can relend.

1. Anticipating SOEs will relend, the bank has no incentive to serve PEs.
2. Relending can always improve social welfare, and all three parties involved are better off.

Part (1) shows that it is better for the bank to lend to SOEs rather than lending to PE directly, knowing that SOEs will relend its fund to the PEs. That is because the bank's expected payoff is actually higher by lending to SOEs, which will relend its fund to the PEs and take over the risk by paying back certain return. In other words, by “unofficially” delegating the financial intermediation role to the SOEs, the bank shifts the risk burden to the SOEs. The intuition of part (2) is the same as that of Proposition 3.

5. Conclusion

In this article, we develop a simple model to explain the puzzles regarding the rapid economic growth in China and its financial repression policy which greatly disfavors PEs. We argue that despite government controls the interest rate and/or the financial resource allocation, SOEs will relend their loans to PEs in the black market. Though maybe illegal, the SOEs acting as financial intermediaries is Pareto improving for two reasons: (1) it helps to reallocate the scarce funds to the more efficient segment; and (2) it

²¹ Please see the appendix for the analysis of the low-type SOE case under the title “Case 2”.

helps to take over the investment risks, which increases the bank's profitability.

Our paper touches two important research areas which have not been paid enough attention so far in the existing literature: the black market for capital in China, and the dynamics in the capital reallocation between the SOEs and the PEs.

People know there exist black markets for capital in China, which is a rational response to the financial repression. The black markets have improved the efficiency of financial resources allocation and supported the fast growth of the private sector. However, due to their illegal nature and thus unavailability of the data, there are few theoretical and empirical works on the subject. Most existing literature (for example, Song et al., 2011) assume there is no capital flow between the SOEs and the PEs in China. However, in reality, there exist enormous amount of capital reallocations between these two sectors through many different channels, such as, the entrusted loans, the trust loans, the trade credits, and many others.

As a first step, this paper identifies the leakage effect and shows that the bank, the SOEs, and the PEs are all participants in the black capital market, which improve the efficiency of the resource allocation despite financial repression. We urge more studies, both theoretical and empirical, should be done in these areas.

There are also interesting extensions on our simple model. For example, we can use it study the relationship between corruption and growth after some straightforward modification; we can build the leakage effect into the growth model to analyze the growth and the dynamics between the state sector and the non-state sector.

Appendix

Proof. (Lemma 1). Given $r_i = \bar{r}_{iH}$, the high-type firm is break even. Moreover, since $\phi_i < \varphi_i(\bar{r}_{iH})$, the low-type firm achieves return of $\pi_{iL}(\bar{r}_{iH}) = \varphi_i(\bar{r}_{iH}) - \phi_i > 0$. Since firms' return is decreasing on interest rate, we have $\bar{r}_{iH} < \bar{r}_{iL}$.

Proof. (Lemma 2). By charging $r_i = \bar{r}_{iH}$, both type firms can borrow, and the bank achieves return $\alpha_i \bar{R}_{iH} + (1 - \alpha_i)[\bar{R}_{iH} - \varphi_i(\bar{r}_{iH})] = \bar{R}_{iH} - (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$. By charging higher interest rate $\bar{r}_{iL}$, only low-type firms can afford to borrow. The bank has return of $\bar{R}_{iL}$. Hence, the bank will charge interest rate $r_i = \bar{r}_{iH}$ if and only if $\bar{R}_{iH} - (1 - \alpha_i)\varphi_i(\bar{r}_{iH}) > \bar{R}_{iL}$, which is equivalent to $\phi_i > (1 - \alpha_i)\varphi_i(\bar{r}_{iH})$.

Case 1. With both interest rate and allocation controls, suppose that it is the low-type SOE that obtains the loan and then relends it to PEs.

When $\phi_B \leq (1 - \alpha_B)\varphi_B(\bar{r}_{BH})$, the SOE only relends to low-type PEs and achieves revenue $\bar{R}_{BL}$. The relending occurs if:

$$\begin{aligned} \bar{R}_{BL} - \bar{r}_{AH} &= \bar{R}_B - \hat{R}_A - \left[\alpha_B \phi_B + (1 - \alpha_A)\phi_A + \int_{-1}^{\bar{r}_{AH}} F_{AH}(R) dR \right] \\ &> \varphi_A(\bar{r}_{AH}) - \phi_A, \end{aligned}$$

where $\varphi_A(\bar{r}_{AH}) - \phi_A$ is low-type SOE's return by keeping the fund for its own project.

When $\phi_B > (1 - \alpha_B)\varphi_B(\bar{r}_{BH})$, the SOE charges low interest rate $\bar{r}_{BH}$ and achieves revenue $\bar{R}_{BH} - (1 - \alpha_B)\varphi_B(\bar{r}_{BH})$. The relending occurs if:

$$\begin{aligned} \bar{R}_{BH} - (1 - \alpha_B)\varphi_B(\bar{r}_{BH}) - \bar{r}_{AH} \\ &= \bar{R}_B - \hat{R}_A - \left[(1 - \alpha_B)(\varphi_B(\bar{r}_{BH}) - \phi_B) + (1 - \alpha_A)\phi_A + \int_{-1}^{\bar{r}_{AH}} F_{AH}(R) dR \right] \\ &> \varphi_A(\bar{r}_{AH}) - \phi_A. \end{aligned}$$

In summary:

1. When $\phi_B \leq (1 - \alpha_B)\varphi_B(\bar{r}_{BH})$, relending is profitable and will occur if $[\varphi_A(\bar{r}_{AH}) - \phi_A] + \left[\alpha_B \phi_B + (1 - \alpha_A)\phi_A + \int_{-1}^{\bar{r}_{AH}} F_{AH}(R) dR \right] < \bar{R}_B - \hat{R}_A$, which can improve social welfare.
2. When $\phi_B > (1 - \alpha_B)\varphi_B(\bar{r}_{BH})$, relending is profitable and will occur if $[\varphi_A(\bar{r}_{AH}) - \phi_A] + \left[(1 - \alpha_B)(\varphi_B(\bar{r}_{BH}) - \phi_B) + (1 - \alpha_A)\phi_A + \int_{-1}^{\bar{r}_{AH}} F_{AH}(R) dR \right] < \bar{R}_B - \hat{R}_A$, which can improve social welfare.

Proof. (Proposition 4) Part (1) has been shown. For part (2), when there is no government control, the bank lends to SOEs if and only if $\bar{R}_B - \hat{R}_A < X$. When there is interest rate control, the bank lends to SOEs if and only if $\bar{R}_B - \hat{R}_A < X + Y$. Since $Y > 0$, it is obvious that the interest rate control will reverse the bank's lending decision and thus will affect social welfare when $(\bar{R}_B - \hat{R}_A) \in (X, X + Y)$. That is, in this region, the bank prefers PEs when there is no control, but prefers SOEs when there is interest rate control. Therefore, we only need to focus on this region to find out the welfare effect of interest rate control.

If $X < X + Y < 0$, then $\bar{R}_B - \hat{R}_A < 0$ when $(\bar{R}_B - \hat{R}_A) \in (X, X + Y)$, and lending to SOEs is social optimal. Therefore, interest rate control improves social welfare.

If $0 < X < X + Y$, then $\bar{R}_B - \hat{R}_A > 0$ when $(\bar{R}_B - \hat{R}_A) \in (X, X + Y)$, and lending to PEs is social optimal. Therefore, interest rate control reduces social welfare.

If $X < 0 < X + Y$, then lending to SOEs is social optimal when $(\bar{R}_B - \hat{R}_A) \in (X, 0)$; lending to PEs is social optimal when $(\bar{R}_B - \hat{R}_A) \in (0, X + Y)$. Therefore, interest rate control improves social welfare when $(\bar{R}_B - \hat{R}_A) \in (X, 0)$, but reduces social welfare

when $(\hat{R}_B - \hat{R}_A) \in (0, X + Y)$.

In summary, interest rate control improves social welfare when $(\hat{R}_B - \hat{R}_A) \in (X, \min\{0, X + Y\})$. It reduces social welfare when $(\hat{R}_B - \hat{R}_A) \in (\max\{0, X\}, X + Y)$.

Case 2. With the interest rate control, suppose that it is the low-type SOE that obtains the loan and then relends to PEs.

When $\phi_B \leq (1 - \alpha_B)\varphi_B(\bar{r}_{BH})$, the relending return is $(\tilde{R}_{BL} - \tilde{R}_{AH}) - \int_{-1}^{\bar{r}_{AH}} F_{AH}(R) dR$. The SOE is willing to relend if $\tilde{R}_{BL} - \tilde{R}_{AH} - \int_{-1}^{\bar{r}_{AH}} F_{AH}(R) dR > \varphi_A(\bar{r}_{AH}) - \phi_A$, where $\varphi_A(\bar{r}_{AH}) - \phi_A$ is low-type SOE's return by keeping the fund for its own project.

When $\phi_B > (1 - \alpha_B)\varphi_B(\bar{r}_{BH})$, the relending return is $(\hat{R}_B - \tilde{R}_{AH}) - (1 - \alpha_B)(\varphi_B(\bar{r}_{BH}) - \phi_B) - \int_{-1}^{\bar{r}_{AH}} F_{AH}(R) dR$. Therefore, the SOE is willing to relend if $\hat{R}_B - \tilde{R}_{AH} - (1 - \alpha_B)(\varphi_B(\bar{r}_{BH}) - \phi_B) - \int_{-1}^{\bar{r}_{AH}} F_{AH}(R) dR > \varphi_A(\bar{r}_{AH}) - \phi_A$.

Knowing SOE is willing to relend, by financing SOEs, the bank's expected return becomes $\bar{r}_{AH}$. By lending to PEs, the expected return of bank can be shown as $\bar{r}_{AH} - \left[\alpha_B \int_{-1}^{\bar{r}_{AH}} F_{BH}(R) dR + (1 - \alpha_B) \int_{-1}^{\bar{r}_{AH}} F_{BL}(R) dR \right] < \bar{r}_{AH}$. Therefore, given the bank anticipates that the SOEs will relend, it has no incentive to serve PEs.

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